

FSP Valuation Framework: Assumptions Pack

Last updated: Apr 2025

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- **Summary of key assumptions**

- Discount rate
- PTRS
- Clinical trial parameters
- Commercial assumptions (US)
- Commercial cost structure
- BD deal terms
- Company exit opportunities
- Biotech program reference

How to use this document



Description

This document provides **benchmarks of key assumptions** that should be used for any and all valuation modeling at Flagship. The data provided within this document is meant as a common starting point to better evaluate assumptions and related metrics as part of decision-making activities. The goal is not to enforce the same inputs, but instead, provide a strawman that shows how inputs have deviated from a common default set of assumptions

This document will be refreshed and published on a quarterly basis by the Business Analytics team. Please note that this document is not an exhaustive list of available industry data



Instruction

There are eight modules in the Assumptions Pack. For each module:

- Review the default benchmarks and input reference (the default case typically represents the industry average, unless stated otherwise)
- Refer to the attached Indication Assumption Workbook (Excel) for indication-level data, when relevant
- Conduct further research with KOLs, both internal and external, to better triangulate for the specific purpose for valuation modeling
- **Create a table that shows deviations from the default assumptions as part of the final output with rationale for any differences**

The Business Analytics team is available to answer any questions about the Assumptions Pack and valuation modeling—please reach out to Heng and Danielle (hlu@flagshippioneering.com, dhuang@flagshippioneering.com)

Overview of the Assumptions Pack

	Modules							
	1 Discount rates	2 PTRS ¹	3 Clinical trial parameters	4 Commercial assumptions (US)	5 Commercial cost structures	6 BD deal terms	7 Company exit opportunities	8 Biotech program reference
Key metrics	Discount rates, determined by tiering	- Probability of phase transitions by TA, indication and modality - PTRS from phase to approval	- Duration of phase - Recruitment time of clinical trials - Total cost of phase - Cost per patient in trial	- Peak sales by TA and indication - Time to peak sales - Revenue ramps	- COGS (including CMC) - SG&A - CapEx	- Deal value and upfront by modality, therapeutic area, and development stage - Milestone and royalty estimates	- Acquisition total deal value, upfronts, and milestones; by TA and development phase - IPO market cap	- Number of assets by development stage - Number of TAs - Deals / collaborations
Data sources	Flagship internal assessment in collaboration with QLS	Internal assessment, EvaluatePharma	Internal assessment, EvaluatePharma	EvaluatePharma	Internal assessment, EvaluatePharma	Internal assessment, GlobalData, EvaluatePharm, Pitchbook	Curated Acquisition Database, FactSet	Market research
Input type	Fixed assumption	Default assumption	Default assumption	Reference datapoints	Default assumption	Reference datapoints	Reference datapoints	Reference datapoints

Input type definitions:

- **Fixed assumption** – all models must use this assumption
- **Default assumption** – deviations from default assumptions are expected to be explicitly noted
- **Reference datapoints** – model expected to have its own informed view for these assumptions, but reference datapoints are provided to help develop an informed view

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Discount rate



Fixed Assumptions

Stage of asset in model	Single discount rate to be calculated using the below tiering
Preclinical	35%
Phase I	25%
Phase II	20%
Phase III	15%
Commercial	10%



Discount Rate ¹

12%-16%

Note that the tiering discount rate must be used in the valuation model. They **vary as the asset progresses through different development stage**, and altogether blend to an equivalent *flat discount rate in the range of ~ 12-16%*, depending on the stage of the current program

1. Illustrative of what a typical range of discount rates might be. Valuation models must use the stage-specific values on the left

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Probability of technical and regulatory success (PTRS)



Default Assumptions

Stage of asset	PTRS for phase transition	PTRS from phase to approval
DC	47%	5%
IND-enabling study	70%	11%
PI	70%	16%
PII	45%	23%
PIII	54%	50%
Filed	93%	93%

Note that PTRS here only applies to the start of each stage, before the study initiation

PTRS for phase transition: based on TA and indication

Therapeutic Area	PI - PII (%)	PII - PIII (%)	PIII - Filed (%)	Filed - Approved (%)
Blood	77%	51%	75%	98%
Cancer	70%	31%	59%	93%
Cardiovascular	80%	43%	43%	94%
Diabetes	66%	39%	37%	99%
Gastro-intestinal	66%	47%	48%	89%
Hepatic & biliary	66%	35%	35%	93%
HIV & related conditions	59%	36%	50%	99%
Hormone	73%	52%	71%	100%
Immunology	73%	47%	53%	99%
Infections	62%	51%	57%	94%
Metabolic	80%	56%	62%	85%
Miscellaneous	79%	63%	82%	93%
Musculoskeletal	71%	53%	48%	87%
Neurology	74%	39%	50%	93%
Psychiatry	72%	26%	20%	79%
Reproduction	70%	43%	59%	67%
Respiratory	68%	32%	45%	92%
Sensory organs	74%	47%	45%	96%
Skin	75%	46%	64%	98%
Surgery	69%	73%	70%	100%
Urinary tract	60%	37%	31%	96%
Cross-TA average	70%	45%	54%	93%

Therapeutic Area	No. of PI studies	No. of PII studies	No. of PIII studies	No. of Filed studies
Blood	526	509	171	122
Cancer	9816	6512	939	433
Cardiovascular	796	895	258	109
Diabetes	574	488	88	40
Gastro-intestinal	573	523	132	69
Hepatic & biliary	470	273	29	13
HIV & related conditions	214	170	42	30
Hormone	94	80	26	17
Immunology	849	602	156	71
Infections	2112	1717	459	256
Metabolic	371	266	77	41
Miscellaneous	455	285	89	68
Musculoskeletal	705	657	171	94
Neurology	1704	1378	333	151
Psychiatry	480	475	145	46
Reproduction	165	211	58	35
Respiratory	714	649	96	50
Sensory organs	561	550	116	52
Skin	879	814	160	93
Surgery	55	67	23	16
Urinary tract	379	276	53	20
Total	22492	17397	3621	1826

To further triangulate the right PTRS for your model:

- See next page on modality-specific PTRS for your technology
- For indication-specific benchmarks, refer to **Indication Assumption Workbook** (Excel)

Please note that Therapeutic Areas from Evaluate Pharma are categorized based on organ systems, thus Autoimmune Diseases will fall under different Therapeutic Areas (refer to the full list of diseases available in the Indication Assumption Workbook)



PTRS for phase transition: based on modality

Modality	PI to PII	PII to PIII	PIII to Filed	Filed to Approved
Biotechnology				
Cell therapy	58%	28%	55%	100%
DNA & RNA therapeutics	65%	23%	47%	91%
Gene therapy	77%	23%	46%	82%
Gene-Modified Cell Therapy	48%	18%	56%	100%
Genome Editing	73%	38%	100%	100%
Monoclonal antibody	68%	37%	52%	94%
Oncolytic Virus	56%	22%	20%	100%
Other biotechnology product	58%	19%	55%	100%
Protein & peptide therapeutics	65%	39%	53%	96%
Recombinant antibody	82%	47%	70%	98%
Transgenic product		20%	50%	100%
Vaccine	61%	37%	50%	97%
Conventional				
In vivo diagnostics	64%	50%	77%	97%
Miscellaneous	25%			
Plant extract	63%	39%	50%	95%
Plasma-derived therapy	88%	63%	71%	100%
Small molecule chemistry	64%	33%	47%	91%
Cross-modality average	63%	33%	56%	96%

Modality	No. of PI studies	No. of PII studies	No. of PIII studies	No. of Filed studies
Cell therapy	793	542	71	37
DNA & RNA therapeutics	476	315	55	22
Gene therapy	321	220	26	11
Gene-Modified Cell Therapy	838	292	32	15
Genome Editing	26	9	2	2
Monoclonal antibody	3,472	1,874	306	134
Oncolytic Virus	141	60	6	1
Other biotechnology product	180	95	11	6
Protein & peptide therapeutics	1,496	1,265	327	172
Recombinant antibody	967	908	286	191
Transgenic product		5	2	1
Vaccine	1,425	1,109	196	101
In vivo diagnostics	173	138	47	35
Miscellaneous	8	3		
Plant extract	125	159	36	19
Plasma-derived therapy	88	121	76	60
Small molecule chemistry	11,724	10,278	2,144	1,023

To further triangulate the right PTRS for your model:

- See previous page on TA-specific PTRS for potential disease area
- For indication-specific benchmarks, refer to **Indication Assumption Workbook (Excel)**



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Clinical trial parameters



Default Assumptions

Development stage	Duration	Recruitment time	Trial cost per patient	Total cost of stage
Lead ID	16 months	N/A	N/A	\$5M
Lead Op	12 months			\$10M
DC	12 months			\$15M
PI	24 months	9 months	\$120K	\$4M / \$19M (median / mean)
PII	37 months	15 months	\$130K	\$10M / \$36M (median / mean)
PIII	42 months	17 months	\$130K	\$58M / \$238M (median / mean)
Filed (US)	11 months	N/A	N/A	\$2.5M

- Median values of preclinical / clinical trial parameters on a per-program basis are listed above (unless stated otherwise), excluding all platform-related expense
- Mean values of preclinical / clinical costs are recommended, as they are time adjusted, and most new FSP platform programs focus on challenging diseases that would require higher budget

1. Phase I to Filed—based on Evaluate Omnium, a ML tool that analyzes large data sets across the Evaluate Pharma database; data here are based on NMEs
2. Preclinical duration and cost—use PM's standard assumptions

Clinical trial parameters: phase duration by TAs

Therapeutic Area	Median Filing Time (Months)	Median Filing Time in EMA (Months)	Median Time in PIII (Months)	Median Time in PII (Months)	Median Time in PI (Months)
Blood	11	14	47	38	20
Cancer	8	13	38	36	30
Cardiovascular	16	13	49	36	22
Diabetes	12	13	33	30	23
Gastro-intestinal	13	13	42	36	25
Hepatic & biliary	8	14	47	49	33
HIV & related conditions	11	14	42	32	23
Hormone	14	16	58	46	30
Immunology	9	12	42	32	31
Infections	11	13	41	38	21
Metabolic	11	14	42	35	16
Miscellaneous	12	14	38	29	13
Musculoskeletal	11	13	45	41	22
Neurology	12	14	38	40	21
Psychiatry	17	15	42	33	30
Reproduction	12	14	50	57	12
Respiratory	13	14	43	43	21
Sensory organs	8	11	52	34	33
Skin	13	13	41	36	20
Surgery	15	16	36	35	18
Urinary tract	10	12	35	38	46
Cross-TA average	11	13	42	37	24

For indication-specific benchmarks, refer to **Indication Assumption Workbook** (Excel)

Please note that Therapeutic Areas from Evaluate Pharma are categorized based on organ systems, thus Autoimmune Diseases will fall under different Therapeutic Areas (refer to the full list of diseases available in the Indication Assumption Workbook)

Clinical trial parameters: recruitment time by TAs

Therapeutic Area	Median Recruitment Time in PIII (Months)	Median Recruitment Time in PII (Months)	Median Recruitment Time in PI (Months)
Blood	17	19	16
Cancer	23	28	26
Cardiovascular	29	17	5
Diabetes	19	12	4
Gastro-intestinal	15	13	11
Hepatic & biliary	24	16	11
HIV & related conditions	10	10	8
Hormone	20	12	5
Immunology	30	25	14
Infections	10	10	7
Metabolic	15	13	7
Miscellaneous	13	16	15
Musculoskeletal	17	12	10
Neurology	15	16	8
Psychiatry	16	14	5
Reproduction	12	9	4
Respiratory	16	14	7
Sensory organs	16	13	10
Skin	13	9	8
Surgery	6	25	5
Urinary tract	24	18	9
Cross-TA average	17	15	9

For indication-specific benchmarks, refer to **Indication Assumption Workbook** (Excel)

Please note that Therapeutic Areas from Evaluate Pharma are categorized based on organ systems, thus Autoimmune Diseases will fall under different Therapeutic Areas (refer to the full list of diseases available in the Indication Assumption Workbook)

Clinical trial parameters: cost per patient by TAs

Therapeutic Area	PIII Cost per Patient Median (\$)	PII Cost per Patient Median (\$)	PI Cost per Patient Median (\$)
Blood	\$282,547	\$216,503	\$230,205
Cancer	\$254,146	\$223,784	\$232,836
Cardiovascular	\$97,899	\$159,900	\$102,284
Diabetes	\$120,630	\$115,072	\$90,008
Gastro-intestinal	\$82,954	\$85,127	\$91,214
Hepatic & biliary	\$84,678	\$86,103	\$147,893
HIV & related conditions	\$208,364	\$173,977	\$149,550
Hormone	\$130,130	\$127,953	\$184,667
Immunology	\$211,854	\$191,487	\$192,953
Infections	\$72,996	\$59,189	\$61,900
Metabolic	\$171,522	\$175,031	\$135,647
Miscellaneous	\$439,866	\$404,898	\$212,666
Musculoskeletal	\$218,014	\$179,441	\$151,333
Neurology	\$115,442	\$113,876	\$141,138
Psychiatry	\$94,957	\$103,205	\$97,191
Reproduction	\$54,330	\$59,029	\$81,825
Respiratory	\$149,577	\$169,524	\$155,781
Sensory organs	\$44,448	\$51,419	\$102,601
Skin	\$100,406	\$105,274	\$127,913
Surgery	\$97,898	\$117,826	\$90,494
Urinary tract	\$250,934	\$181,919	\$146,024
Cross-TA average	\$129,295	\$132,299	\$120,011

For indication-specific benchmarks, refer to **Indication Assumption Workbook** (Excel)

Please note that Therapeutic Areas from Evaluate Pharma are categorized based on organ systems, thus Autoimmune Diseases will fall under different Therapeutic Areas (refer to the full list of diseases available in the Indication Assumption Workbook)

Clinical trial parameters: trial cost by TAs (mean)

Therapeutic Area	Total PIII Trial Cost Mean (\$m)	Total PII Trial Cost Mean (\$m)	Total PI Trial Cost Mean (\$m)
Blood	127	28	11
Cancer	219	31	18
Cardiovascular	575	38	10
Diabetes	212	25	9
Gastro-intestinal	129	18	9
Hepatic & biliary	45	13	7
HIV & related conditions	572	37	17
Hormone	68	14	9
Immunology	71	17	11
Infections	175	26	6
Metabolic	230	23	9
Miscellaneous	61	21	8
Musculoskeletal	145	29	13
Neurology	121	19	9
Psychiatry	86	15	6
Reproduction	78	19	6
Respiratory	247	34	10
Sensory organs	61	12	5
Skin	130	14	8
Surgery	55	20	9
Urinary tract	107	17	7
Cross-TA average (time-adjusted)	238	36	19

Mean values of clinical costs are recommended (see Default Assumptions), as they are time adjusted, and most new FSP platform programs focus on challenging diseases that would require higher budget. For indication-specific benchmarks, refer to **Indication Assumption Workbook** (Excel)

Distribution of Clinical Trial Cost (\$M)

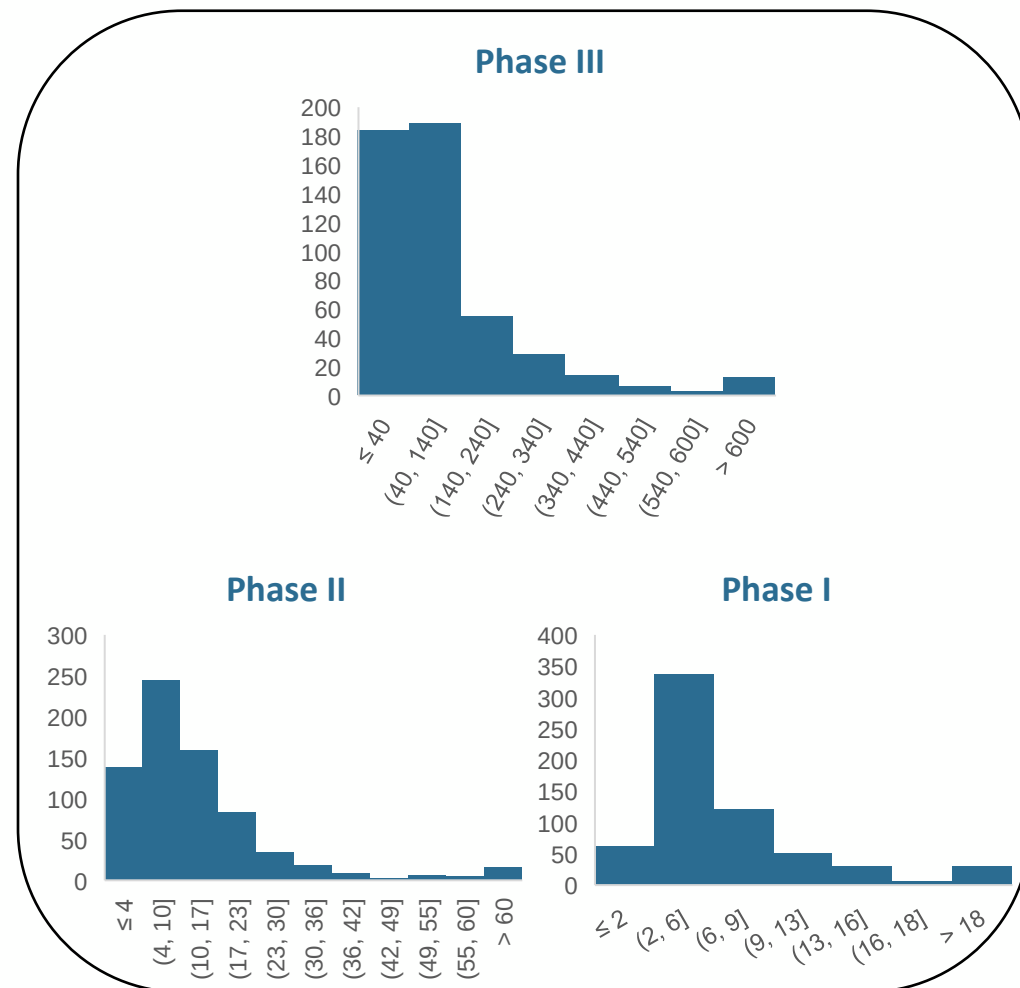


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Commercial assumptions



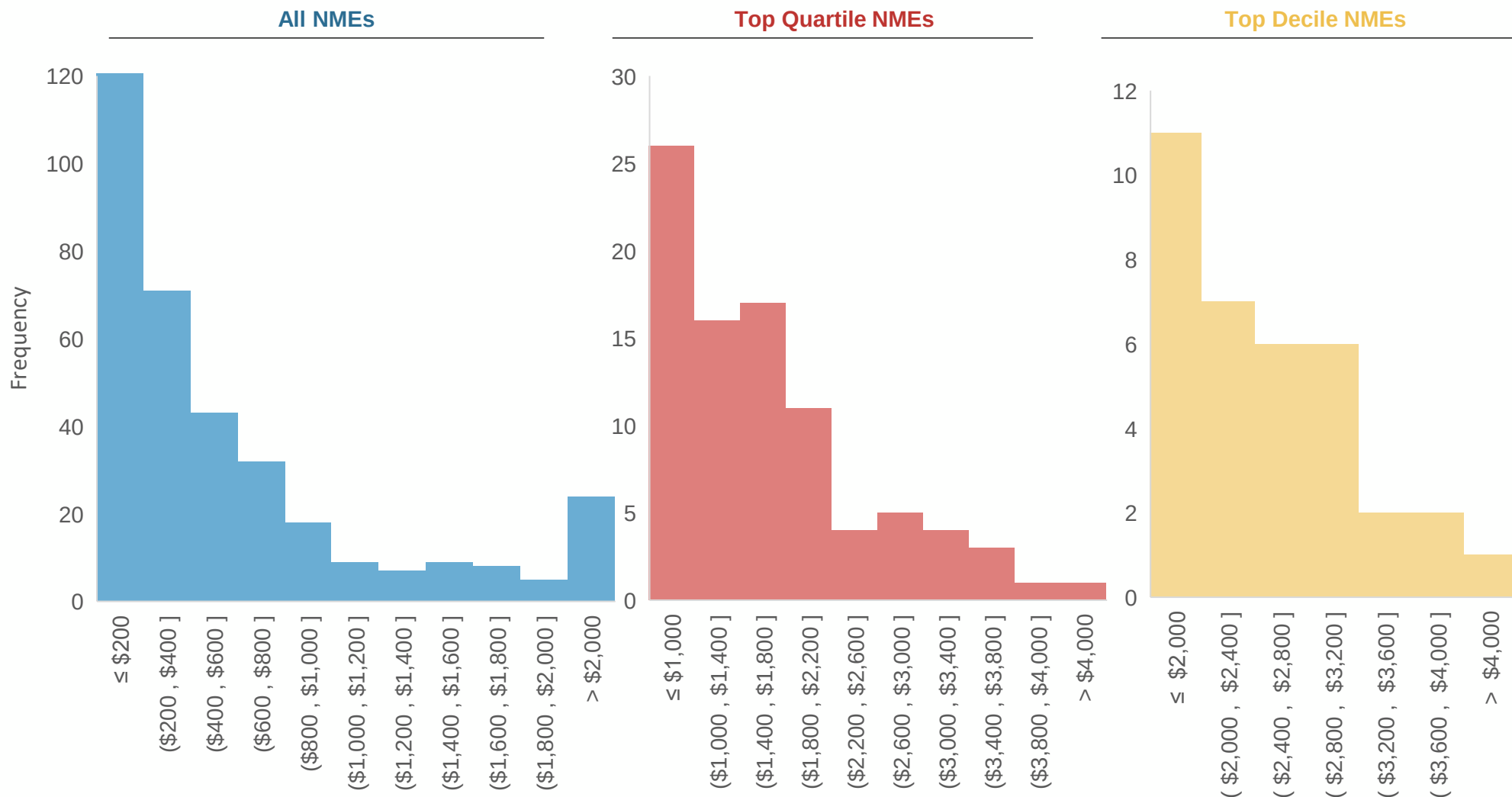
Reference Datapoints

- The industry revenue data in this module are **only for reference**—revenue projections should be performed with extensive market analysis
- Ex-US commercial assumptions** to be discussed on an as-needed basis, after considering global and partnership strategies
 - As a general guideline, ex-US sales are ~1X to 1.75X US sales, with a slower ramp-up

US Peak Sales (time-adjusted)			
	All NMEs	Top quartile NMEs	Top decile NMEs
Peak Sales (median / mean)	\$326M / \$613M	\$1.4B / \$1.6B	\$2.3B / \$2.5B
Time to Peak (median / mean)	9 yr / 9 yr	9 yr / 9 yr	9 yr / 9 yr
Revenue Ramp (US)			
Average commercial life (launch to patent expiry)		13 years	
Sales per Year			
Year since Launch	Growth per Year	% Peak Sale	See Revenue Ramp deep-dive page for Small Molecule and Biologic specific data
1	N/A	7%	
2	217%	23%	
3	66%	38%	
4	31%	49%	
5	21%	60%	
6	14%	68%	
7	15%	78%	
8	14%	89%	
9	12%	100%	
10	-7%	93%	
11	-1%	92%	
12	1%	93%	
13	-7%	87%	

Source: Peak sales are based on Evaluate Omnium, a ML tool that analyzes large data sets across the Evaluate Pharma database; sales are time-adjusted to today's value. Revenue ramp is a data pull from Evaluate Pharma—NME drug products launched between 2010-2014 were analyzed for their annual performance up to 2023 in the US; only NMEs that are small molecules and biologics are included in the analysis

Commercial assumptions: peak sale distributions



- Please note that peak revenue data have a **highly skewed distribution**, where the majority of products are < \$400M
- The industry revenue data in this module are **only for reference**—revenue projections should be performed with extensive market analysis

Commercial assumptions: NME peak sales by TAs

All NMEs

Top Quartile NMEs

Top Decile NMEs

Therapeutic Area	US Peak Sales, Median (\$M)	US Peak Sales, Mean (\$M)	US Peak Sales, Median (\$M)	US Peak Sales, Mean (\$M)	US Peak Sales, Median (\$M)	US Peak Sales, Mean (\$M)	US Median Time to Peak (Yrs)
Blood	259	574	735	1,507	2,595	2,641	7.5
Cancer	338	480	947	1,082	1,319	1,438	8
Cardiovascular	712	980	1,952	2,206	3,114	2,902	8
Diabetes	840	1,220	3,042	3,042	3,042	3,042	10.5
Gastro-intestinal	243	643	2,181	1,915	2,405	2,405	7.5
Hepatic & biliary	303	1,064	3,457	3,457	3,457	3,457	9.5
HIV & related conditions	413	529	1,234	1,234	1,234	1,234	4.5
Hormone	306	792	1,972	1,972	2,807	2,807	11
Immunology	417	480	854	914	1,199	1,199	9.5
Infections	219	513	874	1,487	2,683	2,576	8
Metabolic	307	480	1,167	1,195	1,509	1,509	10.5
Miscellaneous	197	263	571	563	617	617	13
Musculoskeletal	381	849	1,772	2,427	3,488	3,488	9
Neurology	319	626	1,726	1,658	2,317	2,312	9.5
Psychiatry	1,292	1,147	2,248	2,248	2,483	2,483	9
Reproduction	367	551	1,402	1,436	1,623	1,623	7
Respiratory	514	947	2,235	2,235	2,432	2,432	7.75
Sensory organs	326	563	1,698	1,651	1,732	1,732	6.75
Skin	136	564	1,683	1,633	2,348	2,348	9
Surgery	17	487	1,427	1,427	1,427	1,427	9
Urinary tract	494	637	1,143	1,143	1,439	1,439	6.5
Cross-TA average	326	613	1,411	1,662	2,288	2,503	8.5

Source: Peak sales are based on Evaluate Omnium, a ML tool that analyzes large data sets across the Evaluate Pharma database; sales are time-adjusted to today's value. Peak sale data points here represent *indication-level averages* and not *product-level sales*, hence they do not represent the full range of minimum to maximum product sales

Commercial assumptions: NDA peak sales by TAs

NDA (non-NME)

Therapeutic Area	US Median Peak Sales (Time Adjusted) (\$M)	US Median Time to Peak (Years)
Blood	76	11
Cancer	212	10.25
Cardiovascular	204	5.25
Diabetes	311	7.5
Gastro-intestinal	68	6
Hepatic & biliary	1,279	18
HIV & related conditions	1,604	5
Hormone	139	11
Immunology	203	8
Infections	107	6.75
Metabolic	174	7
Miscellaneous	41	7
Musculoskeletal	200	7
Neurology	176	7.5
Psychiatry	295	7
Reproduction	205	8.75
Respiratory	380	8
Sensory organs	126	8
Skin	161	6.25
Surgery	182	2
Urinary tract	465	6
Cross-TA average	180	7.5

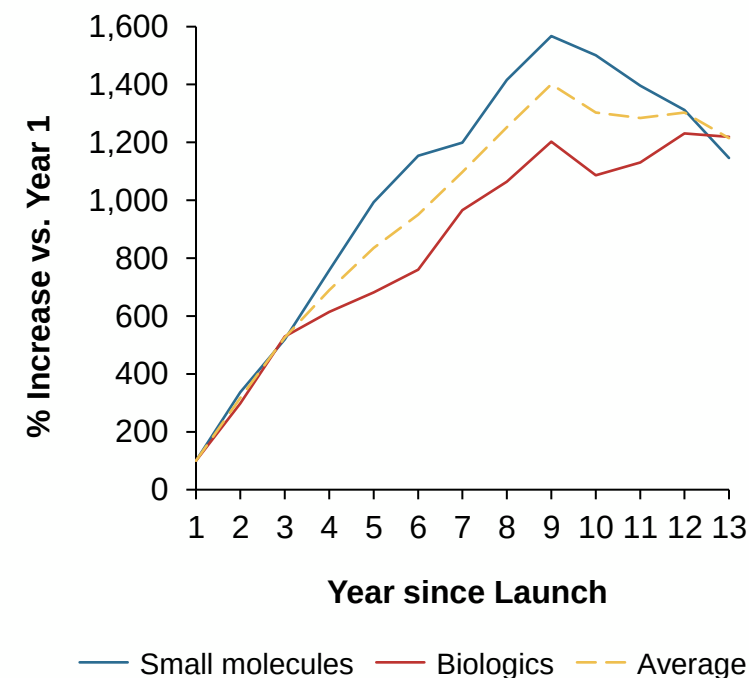
For indication-specific benchmarks, refer to **Indication Assumption Workbook** (Excel)

Definitions

- NME refers to a drug containing an active ingredient that has **never been approved** in the US
- NDA refers to a **non-NME drug** (i.e., new formulations/combinations of previously approved drugs)

Commercial assumptions: revenue ramp for small molecules and biologics

Year since launch	Revenue Growth Per Year			% of Peak Revenue		
	Small molecules	Biologics	Average	Small molecules	Biologics	Average
1	N/A	N/A	N/A	6%	8%	7%
2	237%	198%	217%	21%	24%	23%
3	54%	77%	66%	33%	43%	38%
4	46%	16%	31%	48%	50%	49%
5	31%	11%	21%	63%	55%	60%
6	16%	12%	14%	74%	62%	68%
7	4%	27%	15%	77%	79%	78%
8	18%	10%	14%	90%	86%	89%
9	11%	13%	12%	100%	98%	100%
10	-4%	-10%	-7%	96%	88%	93%
11	-7%	4%	-1%	89%	92%	92%
12	-6%	9%	1%	84%	100%	93%
13	-13%	-1%	-7%	73%	99%	87%



Source: Evaluate Pharma for historical sales analysis—NME drug products launched between 2010-2014 were analyzed for their annual performance up to 2023 in the US; only NMEs that are small molecules and biologics are included in the analysis

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Cost structures



Default Assumptions

Asset-level reference

Modality	COGS (% revenue)	SG&A (% revenue)
Small molecule	10% ¹ (5% for cost-effective processes, potentially in later years)	25% ¹
Biologics	20% ¹	
Gene and Cell Therapy	35% ²	

Company-level reference ³

Company size	COGS (% revenue)	SG&A (% revenue)	CapEx (% revenue)
Large cap (\$30B+)	32%	27%	5%
Mid cap (\$2.5-30B)	35%	30%	5%
Small cap (\$.25-2.5B)	30%	39%	7%

- For finding the right cost structure, determine if the asset or company level input makes the most sense for your modeling needs, and then triangulate COGS, SG&A and CapEx inputs across modality requirements and resourcing needs

- For simplification of tax-related calculations, assume **21% of positive EBIT** when the assets are on the market (excluding partnership R&D milestones or reimbursement)

Source: (1) LEK NPV standard assumptions (2021), Financial ecosystem of pharmaceutical R&D; (2) FSP KOL interviews; (3) Evaluate Pharma—2023 income statements of select biotech companies were used to calculate COGS, SG&A, and CapEx as a percentage of their total revenues

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BD deal terms: deal value and upfront



Reference Datapoints

Note that BD deal terms vary significantly depending on the context—please reach out to the FSP BD Team to align on any assumptions to be used in valuation or LRP

Development Stage at Time of Deal	N	Representative Structure	Median Total Deal Value (\$M)	Median Total Upfront (\$M)	Median Deal Value Per Target (\$M)	Median Upfront Per Target (\$M)
Discovery: Target ID Only	15	Licensee leads novel target ID. Licensor options at validated target and takes over discovery.	625	25	268	7*
Discovery: Target + Lead ID	18	Licensee leads novel target ID and lead ID against the potential target. Licensor options at DC or later.	1078	50	300	7*
Discovery: Lead ID Only	178	Licensee leads lead ID against known target from Licensor. Licensor options at DC or later.	1018	42	319	15*
Preclinical	70	Licensor licenses existing preclinical program OR Licensee continues preclinical development and licensor options at Ph I/II.	783	50	496	30
Phase I	21	Licensor licenses existing Phase I program.	1125	83	864	70

Source: BD team

Transformational deals can be 2-3x total. Some median values calculated from fewer cases than reported N as the specific financial term was not disclosed.

* Deals including separate R&D funding would typically involve lower upfront.

BD deal terms: Target ID, Lead ID, and Preclinical

	TARGET ID		LEAD ID		Preclinical
	Market Deal Range	FSP Company Deal Range	Market Deal Range	FSP Company Deal Range	Market Deal Range
Total Upfront & Milestones per Target (\$)	200M – 350M	240M – 400M	300M - 600M	380M - 600M	350M – 750M
Upfront per Target (\$)	3M – 15M	5M – 15M	5M – 20M	10M - 25M	25M – 60M
R&D Funding (\$)	In Upfront and Milestones	Varies	In Upfront and Milestones	Varies	Varies
Royalties	Low- to high-single digit	Low- to high-single digit	Mid-single to low-double digit	Mid-single to low-double digit	Varies Low-single digit to mid-teens

For discovery deals (target ID, lead ID), the general assumptions for milestones are:

- 45% total deal value is for **Development Milestones**, from discovery through regulatory approval stages
- 55% total deal value is for **Commercial Milestones**

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Company exit opportunities



Reference Datapoints

Value at acquisition ¹	
Median total deal value	\$667M
Median upfront	\$397M
Median milestones	\$300M
Value post-IPO ²	
Median IPO market cap	\$601M

1. Data from 1/1/24 to 11/30/24 in the Business Analytics Acquisition Database, focused on platform/therapeutic companies that have been acquired with an upfront value of >\$25M and can generate immediate liquidity for investors
2. Data from 2024 BioPharma IPO Tracker—FactSet and analysis by Capital Formation team

Company exit opportunities: acquisition value by TA

Therapeutic Area	Median Total Deal Value (\$m)	Median Upfront (\$m)	Median Milestones (\$m)
Autoimmune	629	400	388
Cancer	575	394	409
Cardiovascular	540	415	500
Dermatologic	705	561	252
Endocrine / Metabolic	831	310	320
Gastrointestinal	1,000	1,000	244
Genitourinary	954	500	750
Hematologic	1,100	926	45
Hepatic	565	152	75
Infectious	425	386	100
Inflammation	721	660	400
Musculoskeletal	1,155	860	225
Neurologic	755	307	339
Ophthalmic	423	377	133
Pulmonary	499	206	361
Renal	494	331	300
Transplant	1,600	1,600	-
Unknown	304	155	112
Cross-TA average	667	397	300

See next page on specific deal values of acquisition for each phase

7 Company exit opportunities: acquisition value by development phase

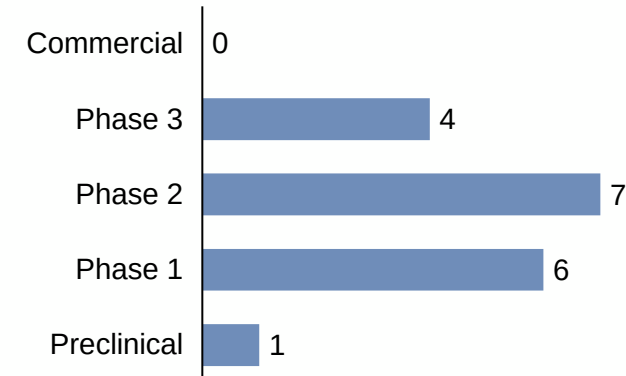
Development phase	Median Total Deal Value (\$m)	Median Upfront (\$m)	Median Milestones (\$m)
Pre-Clinical	468	150	450
Phase I	430	130	380
Phase II	629	320	400
Phase III	1,399	1,150	221
Approved	2,750	2,250	118
Cross-phase average	629	320	380

See previous page on specific deal values of acquisition for each therapeutic area

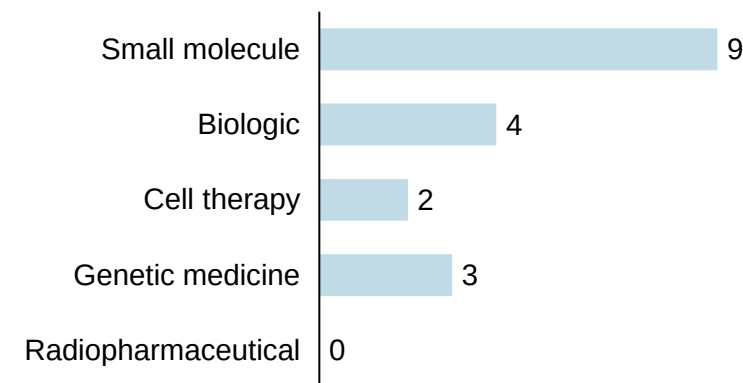
Company exit opportunities: 2024 BioPharma IPO

Biopharma IPOs	2024		2023	
	Average	Median	Average	Median
Count	18		12	
Market Cap	\$651	\$601		
Pre-\$ Equity Value	\$442	\$458	\$728	\$423
Step-Up (x)	1.1	1.1	1.3	1.4
IPO Size	\$204	\$175	\$231	\$228
1-Day Performance	13.9%	9.0%	5.5%	-1.7%
Performance (pricing date to current)	-13.6%	-4.2%	10.7%	-25.2%

Stage of Company at IPO



Primary Modality of Company at IPO



7 Company exit opportunities: BioPharma IPO Tracker

			Pre-\$ Equity					Current market	1-Day			Performance			
Pricing Date	Company	Ticker	Value	Step-Up (x)	IPO Size	Market Cap	% of Co. Sold	cap (\$M)	Performance	IPO Price	Current Price	(pricing date to current)	Company Stage	Therapeutic Area	Primary modality
2/2/2023	Structure Tx	GPCR-US	\$413	1.2	\$185	\$573	32.3%	\$1,943	73.3%	\$15.0	\$33.9	126.3%	Phase 1	Endocrinology	Small molecule
2/9/2023	Mineralys Tx	MLYS-US	\$433	1.7	\$221	\$625	35.3%	\$671	15.3%	\$16.0	\$13.5	-15.7%	Phase 2	Cardiovascular	Small molecule
5/4/2023	Acelyrin	SLRN-US	\$1,129	1.5	\$621	\$1,770	35.1%	\$448	30.6%	\$18.0	\$4.5	-75.2%	Phase 3	Immunology	Antibodies
7/13/2023	Apogee Tx	APGE-US	\$285	2	\$300	\$809	37.1%	\$2,700	24.9%	\$17.0	\$46.1	171.4%	Preclinical	Immunology	Antibodies
7/13/2023	Sagimet Bio	SGMT-US	\$271	1.5	\$85	\$355	23.9%	\$173	-0.3%	\$16.0	\$5.4	-66.5%	Phase 3	Immunology	Small molecule
7/21/2023	Turnstone Bio	TSBX-US	\$186	0.5	\$80	\$266	30.1%	\$13	-8.3%	\$12.0	\$0.6	-95.4%	Phase 1	Oncology	Cell therapy
9/14/2023	Rayze Bio	RYZB-US	\$685	1.6	\$358	\$1,094	32.7%	#N/A	23.3%	\$18.0	\$62.5	247.2%	Phase 3	Oncology	Radiopharmaceutical
9/14/2023	Neumora Tx	NMRA-US	\$2,333	1.4	\$250	\$2,583	9.7%	\$1,583	-17.0%	\$17.0	\$9.8	-42.4%	Phase 3	CNS	Small molecule
9/29/2023	Adlai Nortye	ANL-US	\$1,983	1.2	\$58	\$809	7.1%	--	-34.8%	\$23.0	\$2.3	-89.9%	Phase 3	Oncology	Small molecule
10/19/2023	Abivax	ABVX-US	\$544	N/A	\$236	\$782	30.2%	\$537	-29.0%	\$11.6	\$8.5	-26.9%	Phase 3	Immunology	Small molecule
11/2/2023	Lexeo	LXEO-US	\$172	0.6	\$100	\$272	36.8%	\$278	-8.6%	\$11.0	\$8.4	-23.5%	Phase 1	Cardiovascular	Genetic medicine
11/9/2023	Cargo Tx	CRGX-US	\$299	1.1	\$281	\$580	48.4%	\$824	-3.1%	\$15.0	\$17.9	19.3%	Phase 2	Oncology	Cell therapy
1/24/2024	CG Oncology	CGON-US	\$826	1.5	\$437	\$1,206	36.2%	\$2,419	95.6%	\$19.0	\$35.8	88.3%	Phase 3	Oncology	Biologic
1/25/2024	ArriVent Bio	AVBP-US	\$400	1.1	\$175	\$575	30.4%	\$996	11.1%	\$18.0	\$29.6	64.2%	Phase 3	Oncology	Small molecule
2/2/2024	Alto Neuro	ANRO-US	\$280	1.1	\$175	\$575	30.4%	\$118	29.4%	\$16.0	\$4.4	-72.6%	Phase 2	CNS	Small molecule
2/2/2024	Fractyl Health	GUTS-US	\$604	0.8	\$110	\$714	15.4%	\$125	-14.3%	\$15.0	\$2.6	-82.7%	Phase 3	Metabolic	Genetic medicine
2/7/2024	Kyverna Tx	KYTX-US	\$579	1.4	\$319	\$898	35.5%	\$280	36.4%	\$22.0	\$6.5	-70.5%	Phase 2	Autoimmune	Cell therapy
2/8/2024	Metagenomi	MGX-US	\$469	1.3	\$94	\$563	16.7%	\$78	-31.3%	\$15.0	\$2.1	-86.1%	Preclinical	Rare diseases	Genetic medicine
3/27/2024	Boundless Bio	BOLD-US	\$256	1.2	\$100	\$356	28.1%	\$61	-10.9%	\$16.0	\$2.8	-82.8%	Phase 1	Oncology	Small molecule
4/4/2024	Contineum Tx	CTNM-US	\$292	1.1	\$110	\$402	27.4%	\$390	-3.8%	\$16.0	\$15.1	-5.4%	Phase 2	CNS	Small molecule
6/6/2024	Rapport Tx	RAPP-US	\$465	0.7	\$136	\$601	22.6%	\$878	22.4%	\$17.0	\$24.0	41.2%	Phase 1	CNS	Small molecule
6/27/2024	Alumis	ALMS-US	\$620	1.1	\$210	\$830	25.3%	\$450	-16.9%	\$16.0	\$8.3	-48.3%	Phase 2	Immunology	Small molecule
7/19/2024	Artiva	ARTV-US	\$107	0.2	\$167	\$274	60.9%	\$310	0.0%	\$12.0	\$12.8	6.3%	Phase 1	Autoimmune	Cell therapy
9/12/2024	MBX Bio	MBX-US	\$322	0.9	\$163	\$510	32.0%	\$634	47.8%	\$16.0	\$19.0	18.8%	Phase 2	Endocrinology	Small molecule
9/12/2024	Bicara Tx	BCAX-US	\$617	1.4	\$362	\$932	38.9%	\$1,089	30.1%	\$18.0	\$20.0	11.2%	Phase 1	Oncology	Biologic
9/12/2024	Zenas Bio	ZBIO-US	\$451	1.1	\$225	\$677	33.3%	\$468	6.8%	\$17.0	\$11.2	-34.1%	Phase 3	CNS	Biologic
9/25/2024	BioAge	BIOA-US	\$417	1.2	\$227	\$645	35.2%	\$625	1.7%	\$18.0	\$17.5	-3.1%	Phase 2	CNS	Small molecule

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Biotech program reference



Reference Datapoints

On average, Biotechs at IPO and at Maturity have the following characteristics:		
	Biotech at IPO	Mature Biotech
Number of approved products	0	2 (<i>most advanced stage</i>)
Number of Phase 1 programs	1	4
Number of Phase 2 programs	2 (<i>most advanced stage</i>)	3
Number of Phase 3 programs	0	2
Number of therapeutic areas	3	3
Number of out-licensing deals	4	6
% of clinical programs in collaboration	30%	40%
<i>Taken altogether, this suggests the following in order to get to the most advanced stage ¹:</i>		
Number of discovery programs leading to DCs	8	40
Timeline of program development	5 years	13 years

These benchmarks on R&D activities only serve to guide option value calculation, and should not be used to guide R&D activities for Flagship companies

Source: Li et al (2020), Realizing a biotech's potential, Deloitte Insights

1. Calculation of number of discovery programs and timeline used PTRS and clinical trial parameters from Modules 3 and 4